

# Machine Learning Lab – Question 12

## Question

Implement Iris Flower Classification using KNN.

## Code

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score

iris=load_iris()
X_train,X_test,y_train,y_test=train_test_split(
    iris.data,iris.target,test_size=0.2,random_state=42,stratify=iris.target)

model=KNeighborsClassifier(n_neighbors=3)
model.fit(X_train,y_train)
pred=model.predict(X_test)

print("Accuracy:",round(accuracy_score(y_test,pred),4))
print("New flower:",iris.target_names[model.predict([[5.1,3.5,1.4,0.2]])[0]])
```

## Output

```
Accuracy: 0.9667
New flower: setosa
```